V Raj Narayanan

Kochi, Kerala, India
+91 9746740866 — vrajnarayan666@gmail.com — LinkedIn — GitHub

Profile

Final-year Electronics and Communication Engineering undergraduate at VIT-AP with hands-on experience in embedded systems, PCB design, analog electronics, signal processing, computer vision, and hardware-software integration.

Education

Vellore Institute of Technology, Andhra Pradesh 2023 – 2027
B.Tech in Electronics and Communication Engineering Final Year

Rajagiri Christu Jayanthi Public School 2023
Senior Secondary (CBSE)

Technical Skills

Languages: C, C++, Python, Embedded C, Java
Microcontrollers: ESP32, STM32, Arduino, Raspberry Pi
Protocols: UART, SPI, I2C
Hardware: PCB Design, KiCad, Analog Electronics, Circuit Design, Sensors, Audio Electronics, Signal Conditioning
Tools: MATLAB, Keil uVision, Arduino IDE, OpenCV, PaddleOCR

Experience

Maven Silicon
Embedded Hardware System Design Intern

- Gained hands-on experience with embedded systems and semiconductor design workflows through practical technical training and development.

Technical Member – IETE, VIT-AP

- Contributed to technical workshops, electronics activities, and hardware-oriented student projects within the IETE club.

Projects

SPECTER – Smart Probe for Electronic Circuit Testing Evaluation & Routing *PCB Design, Embedded Systems, AI*

- Developing a multi-modal PCB diagnostic system combining current injection, Kelvin measurement, impedance analysis, TDR, and thermal imaging for fault localization.

Digital Pitch Shifter Pedal PCB *ESP32, WM8960, OPA1652, KiCad*

- Designed a compact digital guitar pitch-shifting PCB with audio ADC/DAC, ESP32 processing, analog buffering, OLED display, and control interface.

Bioimpedance Spectrometer *ESP32, Analog Front-End, Signal Processing*

- Designing a multi-frequency bioimpedance measurement system integrating excitation, sensing, analog signal conditioning, and microcontroller-based processing.

OCR-Based Assistive Glasses *ESP32-CAM, OpenCV, PaddleOCR*

- Developed wearable glasses for real-time text recognition using ESP32-CAM, OpenCV, and PaddleOCR with multilingual text extraction.

Smart Safety System for Trains and Autonomous Vehicles *Arduino, Ultrasonic Sensors, Embedded C*

- Developed an embedded obstacle-detection and collision-prevention system using ultrasonic sensing with automated braking and alert mechanisms.

Certifications

Certificate of Completion – Maven Silicon Internship — AWS Academy – Cloud Foundations — MATLAB Programming
NVIDIA – LLM Applications — STM32F4 Embedded Development – Udemy — Embedded C Programming

Relevant Coursework

Digital Electronics — Microprocessors & Microcontrollers — Analog Electronics — Signals & Systems — Embedded Systems

Languages

English, Malayalam, Hindi, Tamil, Spanish